Response to Editor

Manuscript forests-4515777

"The Effect of China's National Forest City Program on Urban Heat Is Not Robustly Identified: Comment on Ma, Geng, and Zhong (2026) and Demonstration of the Use of Coding Agents for Replication Studies"

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I thank the editor for the careful reading of the manuscript and for the opportunity to revise. The editor raises six concerns. I address each in turn below. Revised passages in the manuscript are indicated by page and line number.

Editor Comment 1: Exact Replication on Ma et al.'s Sample

Editor's request: An exact replication on the Ma et al. (2026) sample and outcome — extreme heat days, 243 cities, 2000–2023 — using heterogeneity-robust estimators.

I agree that an exact replication on Ma et al.'s sample would be the most direct response to their finding. Unfortunately, the data required for such a replication are not available to me.

Ma et al. (2026) draw their weather station data from the China Meteorological Administration (CMA), a proprietary network that requires Chinese institutional affiliation to access. Their city-level control variables come from the China City Statistical Yearbook, which is also not available for download. Both I and Ms. Cui contacted the authors directly to request that they share replication data and we had no response. Obtaining the 243-city sample of CMA weather station data is not feasible.

The one step available to make the analysis more comparable to Ma et al. is to adopt their spatial unit. Ma et al. aggregate outcomes at the prefecture level; the original submission used circular buffers of 5, 10, and 20 km radius. I have therefore added prefecture-level boundary extractions to the analysis, using 2005 prefecture boundary polygons matched to each city in our sample. Of our 275 sample cities, 258 are successfully matched to a prefecture polygon; the remaining 17 lie in regions (mainly Xinjiang, Hainan, and parts of Inner Mongolia) not covered by the shapefile and have missing values for prefecture-level outcomes. Prefecture-level results are consistent with the circular-buffer results across all outcomes: temperature estimates are small, positive, and statistically insignificant.

The remaining gap in sample coverage — 99 cities vs. Ma et al.'s 243 — reflects the difference in station network density between GHCN-Daily (the publicly available source) and CMA (the proprietary source used by Ma et al.). This limitation is now documented explicitly in the revised manuscript.

Editor Comment 2: Valid Aggregate Inference for the Station Hot-Days Analysis

Editor's request: A valid aggregate ATT with a properly documented confidence interval for the station hot-days outcome. The previous submission showed the event-study figure but did not report the aggregate ATT.

I have added the aggregate ATT for the station hot-days outcome to the Results section (Section 3, third paragraph). Table 1, column (6) now reports BJS with controls at the station level, using the same Python implementation as columns (1)–(2), and Appendix Figure A4 shows the corresponding event studies.

All estimates are statistically indistinguishable from zero, consistent with the city-level null results and with the paper's overall conclusion that an NFCP temperature effect is not robustly identified.

Editor Comment 3: Treatment of Missing Meteorological Observations

Editor's request: The paper does not describe how station-year observations with few valid daily readings are handled.

I have added the following to the Data and Methods section (Section 2, station data paragraph):

> "We retain stations within 50 km of a city centroid with at least 30 valid daily maximum temperature readings per year; city-years for which no station meets this threshold are excluded from the estimation sample. Where a city has multiple qualifying stations, all outcomes are averaged across stations to yield one observation per city-year."

Editor Comment 4: Title and Conclusion

Editor's request: The title and conclusion are too strong; "not robustly identified" is what the evidence supports, not "no effect."

I agree with this characterisation and have made the following changes.

The title is changed from "No Effect of China's National Forest City Program on Urban Heat or Forest Cover: A Comment on Ma, Geng, and Zhong (2026)" to "The Effect of China's

National Forest City Program on Urban Heat Is Not Robustly Identified: Comment on Ma et al. (2026) and Demonstration of the Use of Coding Agents for Replication Studies." I think that part of the interest of this project lies in its use of coding agents to conduct a replication exercise that would otherwise be impractical, and the revised title now reflects this.

Throughout the rest of the document I tempered the language from "find no statistically or economically significant cooling effect" to "find that a cooling effect of NFCP designation is not robustly identified."

Editor Comment 5: Factual Inaccuracies About Ma et al.

Editor's request: Correction of factual inaccuracies about Ma et al. (2026).

I am not certain about what you are referring to. I have made the following corrections and additions. If I missed the problem you were thinking of, let me know.

1. **Pencil-test claim.** The previous submission stated that "a straight line fits within the confidence intervals of all pre- and post-designation coefficients" in Ma et al.'s event study. This is incorrect: the post-designation coefficients in their Figure 1 reach approximately -8 extreme heat days by period $+15$, with relatively narrow confidence intervals by that point. I have corrected this to note that their event-study plot shows a decreasing pre-trend, suggesting a violation of the parallel trends assumption.
2. **Description of Ma et al.'s estimator.** The previous submission stated that Ma et al. "apply this estimator [BJS] for their event study but not their main results table." I have corrected this to note that their main estimates use a TWFE regression known to be biased under heterogeneous treatment effects (Goodman-Bacon, 2021).

Editor Comment 6: Data and Code Availability

Editor's request: Full data and code availability statement; correction of factual inaccuracies about Ma et al. (2026).

1. **Data and Code Availability.** The revised manuscript lists the public sources for each data layer. It also gives a link to the Git repository containing a replication package.

References

Andrew Goodman-Bacon. Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277, 2021. doi: 10.1016/j.jeconom.2021.03.014.

Yun Ma, Yan Geng, and Shuai Zhong. Urban forest development and extreme heat mitigation: The climate adaptation effects of China's National Forest City Policy. *Forests*, 17 (1):79, 2026. doi: 10.3390/f17010079.